Mecanismos

CURSO 2021-22

Tercer curso GIA

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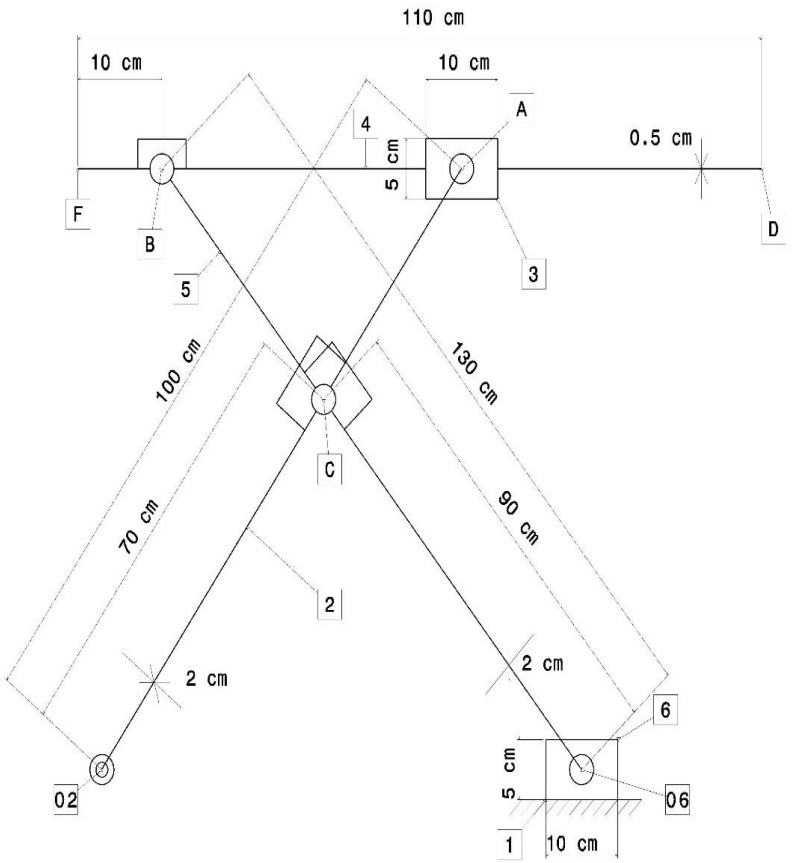
1. INTRODUCCIÓN

1.1. FOTO DEL MECANISMO, ESQUEMA Y BREVE DESCRIPCIÓN DE SU FUNCIONAMIENTO



1.2. JUSTIFICACIÓN DEL NÚMERO DE GRADOS DE LIBERTAD

Ley de Gruebler para el movimiento plano:

$$\boxed{G = 3(N-1) - 2g - h}$$

Particularizando para nuestro mecanismo:

$$G = 3(6-1) - 2 \times 7 = 1$$

1.3.DIMENSIONES, MASAS E INERCIAS DE LAS BARRAS

Masa: $m = \frac{\rho\, V}{1000}$, con ρ = 2.7 g/cm^3

Dados el espesor e, la longitud l y la profundidad p:

- Si la sección es circular: $V = \pi \left(\frac{e}{2}\right)^2 l$
- Si la sección es rectangular: $V = l\, e\, p$

HIPÓTESIS:

- Al se distribuye homogéneamente → G≡punto medio
- El mecanismo realiza un movimiento plano ≡ I_{zz}
- Para simplificar el cálculo de las inercias supondremos que las varillas y placas son delgadas

1.3.DIMENSIONES, MASAS E INERCIAS DE LAS BARRAS

$$\overleftrightarrow{I}(G) = \frac{m\left(\frac{l}{2}\right)^2}{3}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\overleftrightarrow{I}(G) = \frac{m}{3}\begin{bmatrix} \left(\frac{e}{2}\right)^2 & 0 & 0 \\ 0 & \left(\frac{l}{2}\right)^2 & 0 \\ 0 & 0 & \left(\frac{e}{2}\right)^2 + \left(\frac{l}{2}\right)^2 \end{bmatrix}$$

Barras 2 y 5: varillas de sección circular

Barra 4: varilla de sección rectangular

Barras 3 y 6: placas rectangulares

Barras	Dimensiones [l e p] (cm^3)	Masa (kg)	Inercia (kg cm^2)
1	-	-	-
2	100 x 2 x 2	0.84823	706.858
3	10 x 5 x 0.2	0.027	0.28125
4	110 x 0.5 x 30	4.455	4492.125
5	130 x 2 x 2	1.1027	1552.968
6	10 x 5 x 0.2	0.027	0.28125

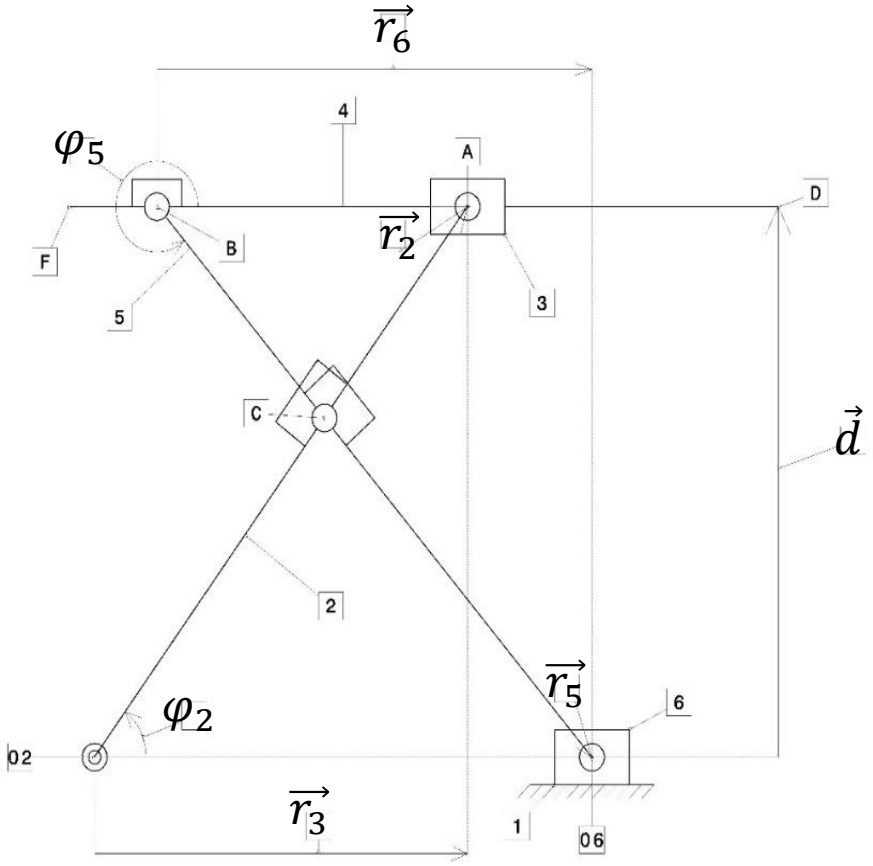
1.4.BARRA DE ENTRADA Y LEY DE MOVIMIENTO

Barra de entrada: 2 → $\varphi_2 = \varphi_{20} + \omega_2 t$

con $\omega_2 = 0.24\, rad/s$

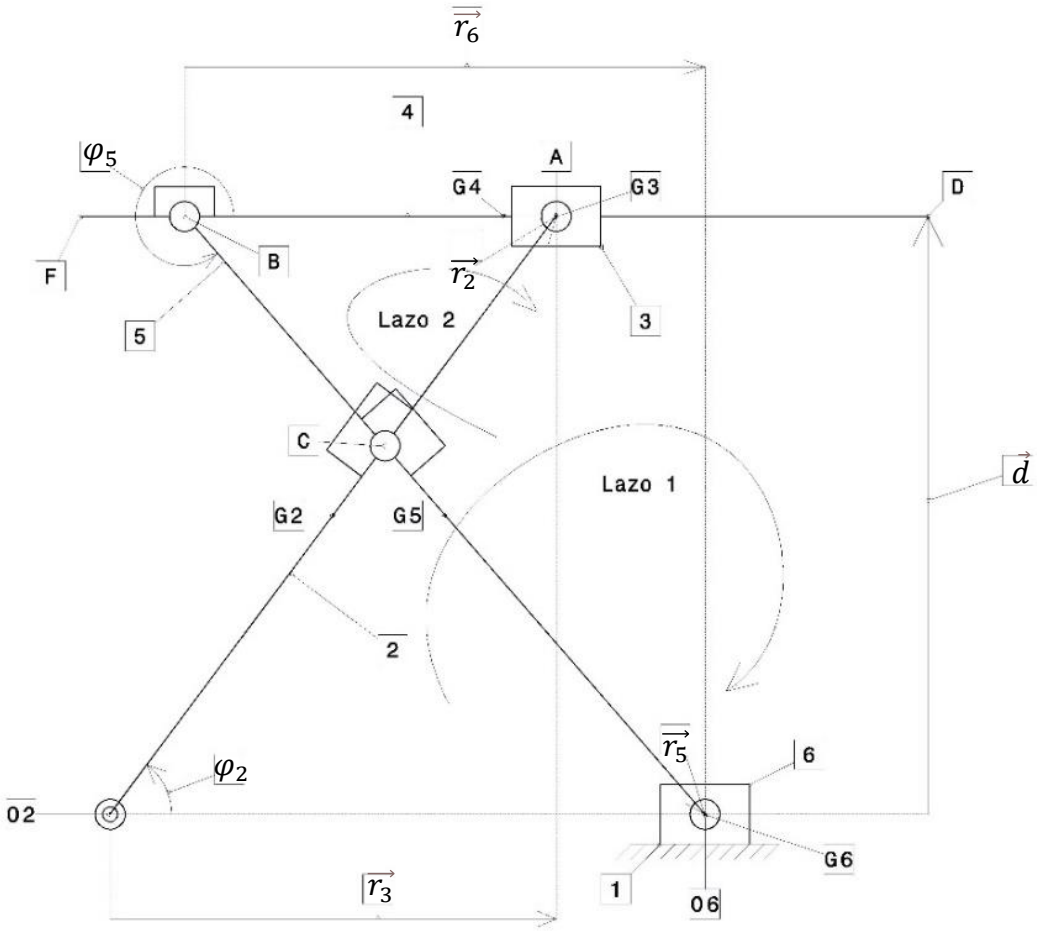
Rango de valores en el que puede describirse el movimiento:

$$\varphi_2 = [0,1.2]\, rad$$
$$r_3 = [100\,, 35]\, cm$$
$$\varphi_5 = [2\pi\,, 5.5\,]\, rad$$
$$r_6 = [130,90]\, cm$$
$$d = [0,93]\, cm$$



2. ANÁLISIS CINEMÁTICO DEL MECANISMO

2.1. ECUACIONES DE LAZO



- Lazo 1: $\vec{r}_2 = \vec{r}_3 + \vec{d}$

  $r_2 cos\varphi_2 = r_3$

  $r_2 sen\varphi_2 = d$

- Lazo 2: $\vec{r}_5 + \vec{d} = \vec{r}_6$

  $r_5 cos\varphi_5 = r_6$

  $r_5 sen\varphi_5 + d = 0$

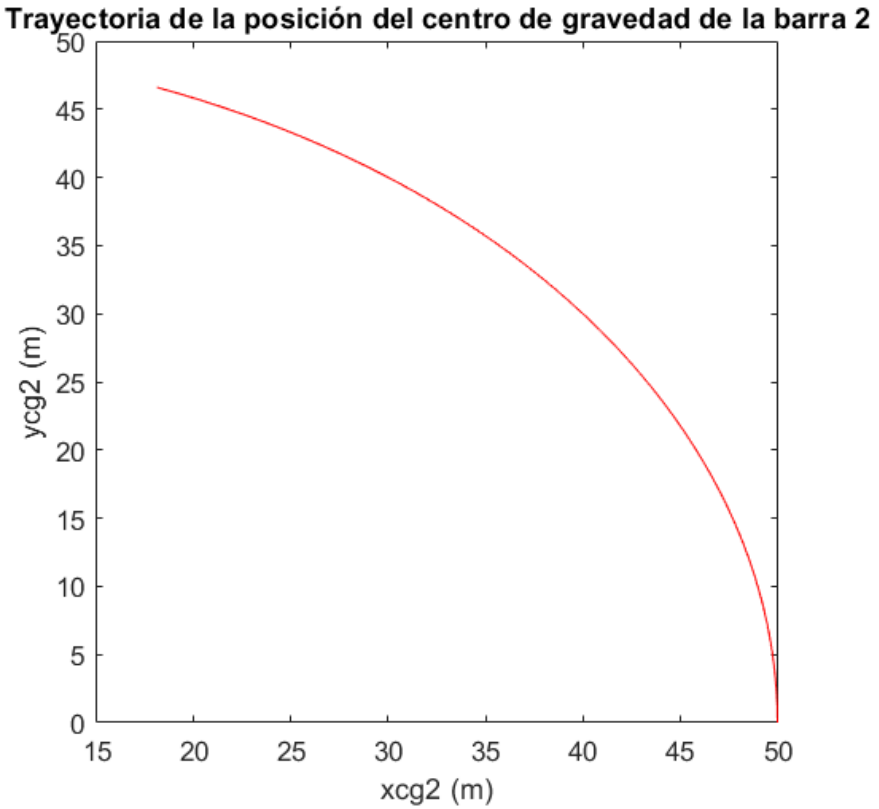
2.2. ANÁLISIS DE LAS POSIBLES CONFIGURACIONES SINGULARES

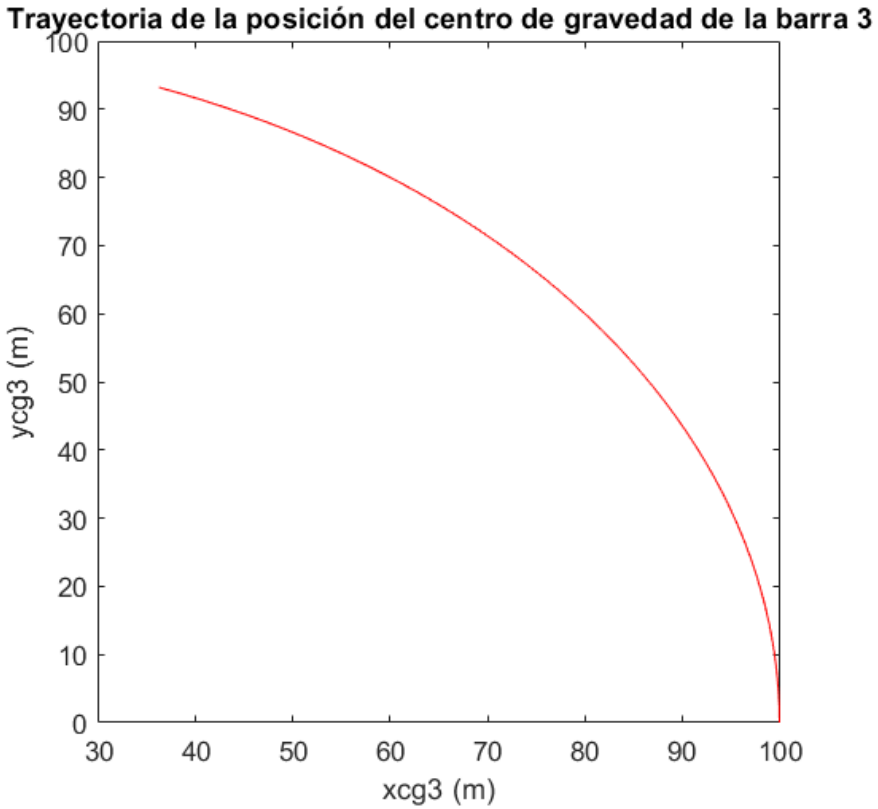
$$J = \begin{bmatrix} -r_2 \operatorname{sen} \varphi_2 & -1 & 0 & 0 & 0 \\ r_2 \cos \varphi_2 & 0 & 0 & 0 & -1 \\ 0 & 0 & -r_5 \operatorname{sen} \varphi_5 & -1 & 0 \\ 0 & 0 & r_5 \cos \varphi_5 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$|J| = 1 \begin{vmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & -r_5 \operatorname{sen} \varphi_5 & -1 & 0 \\ 0 & r_5 \cos \varphi_5 & 0 & 1 \end{vmatrix} = -1 \begin{vmatrix} 0 & 0 & -1 \\ -r_5 \operatorname{sen} \varphi_5 & -1 & 0 \\ r_5 \cos \varphi_5 & 0 & 1 \end{vmatrix} = r_5 \cos \varphi_5$$

No presenta configuraciones singulares

2.3. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DEL CENTRO DE GRAVEDAD Y DE LA COORDENADA ANGULAR DE CADA BARRA





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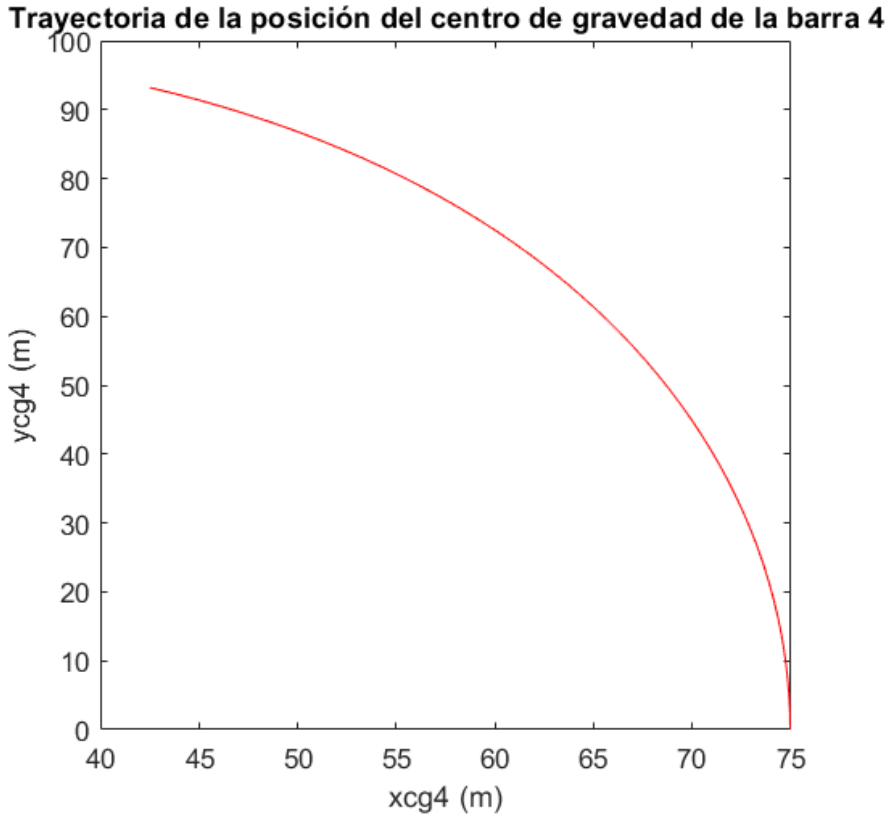

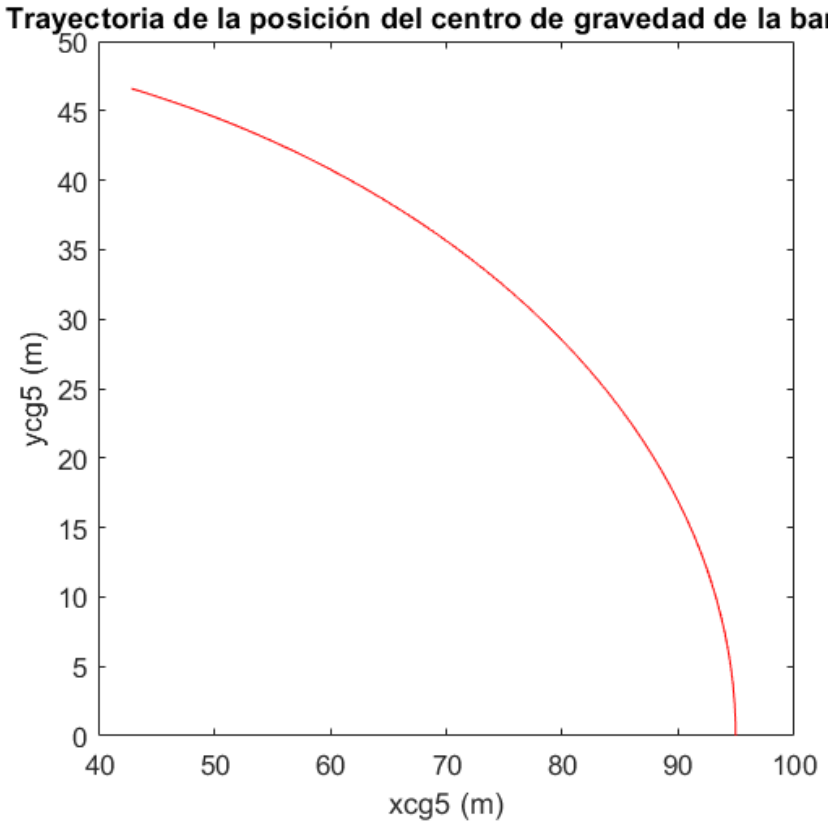

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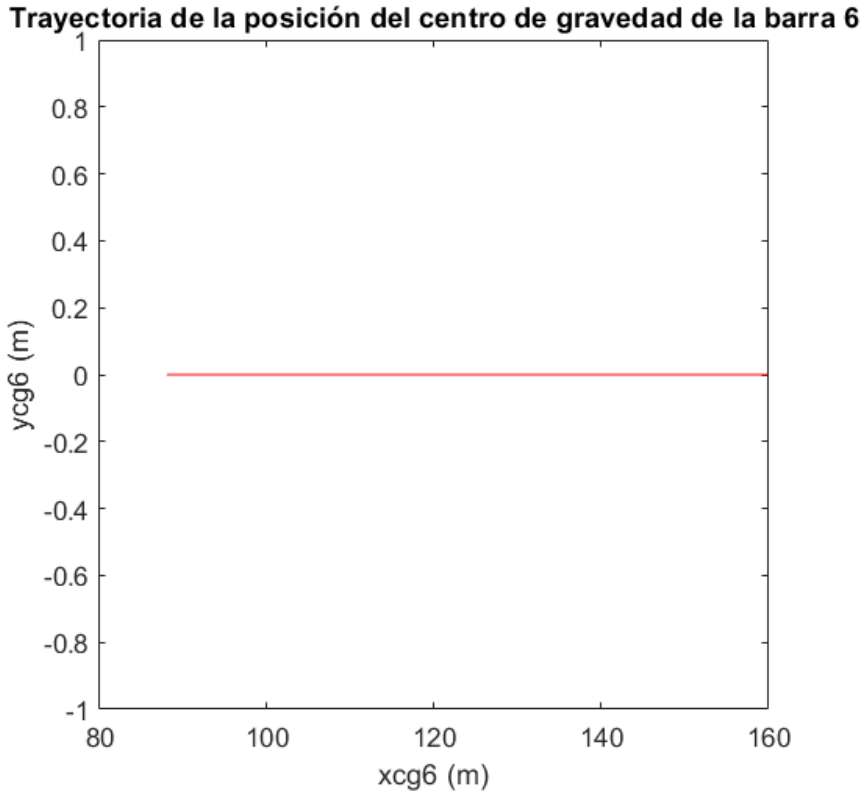

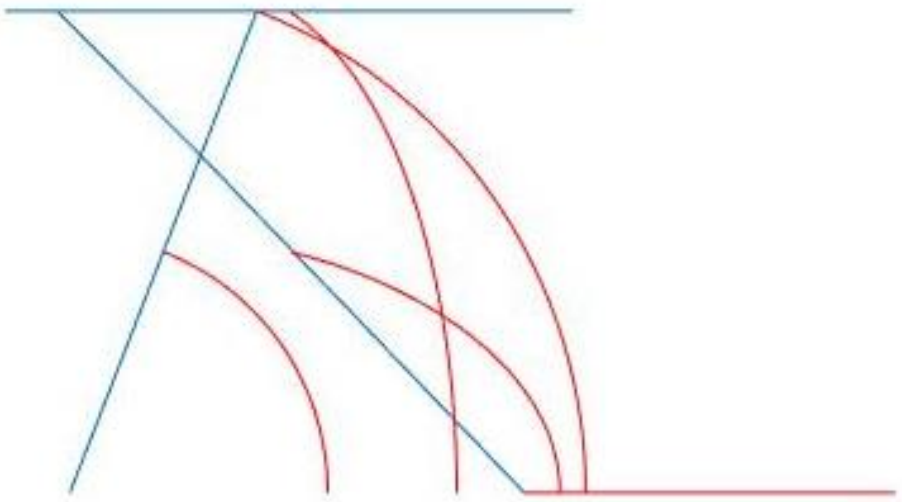

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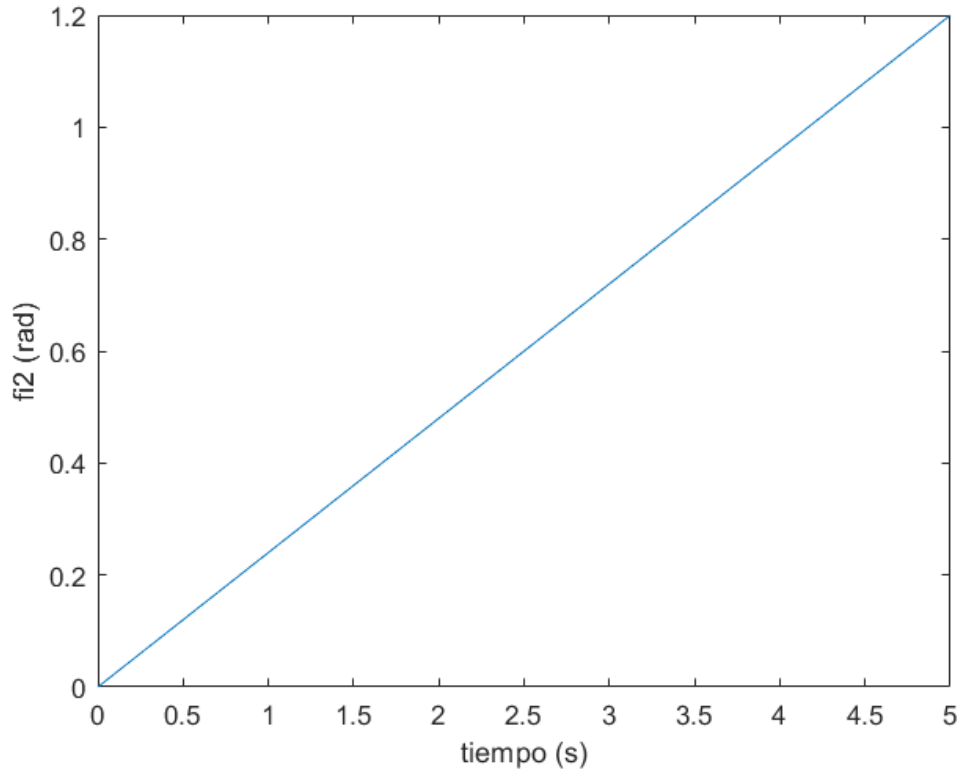

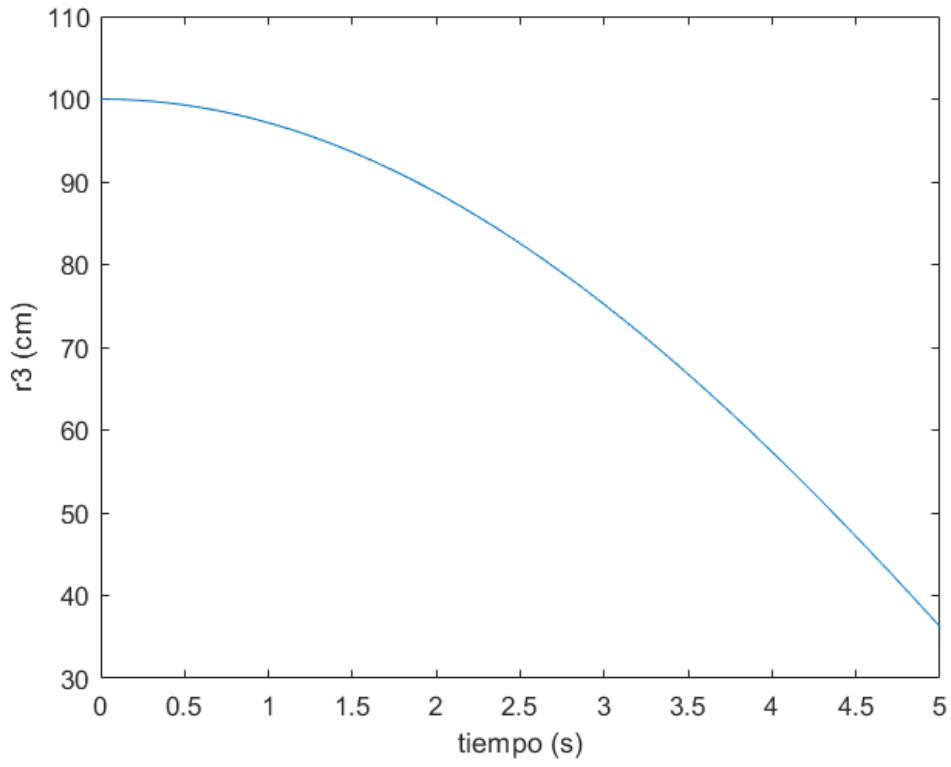

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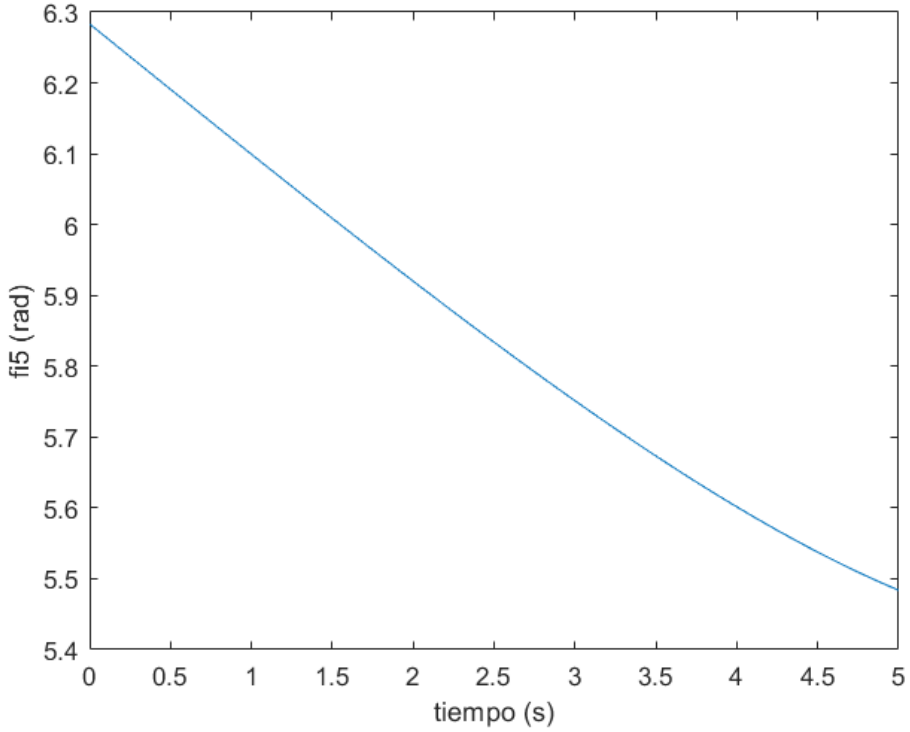

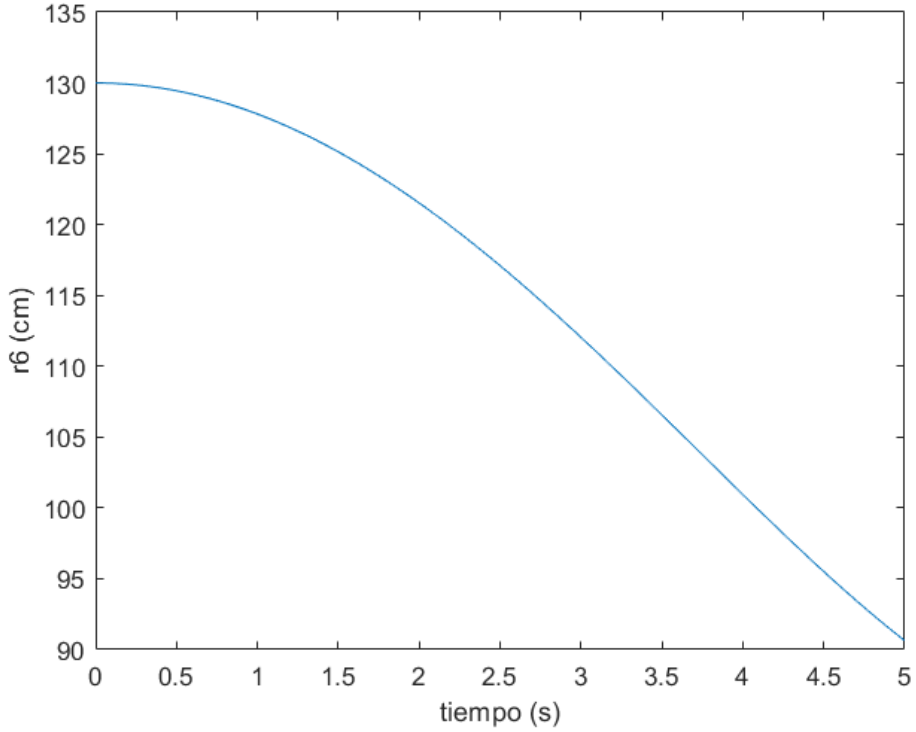

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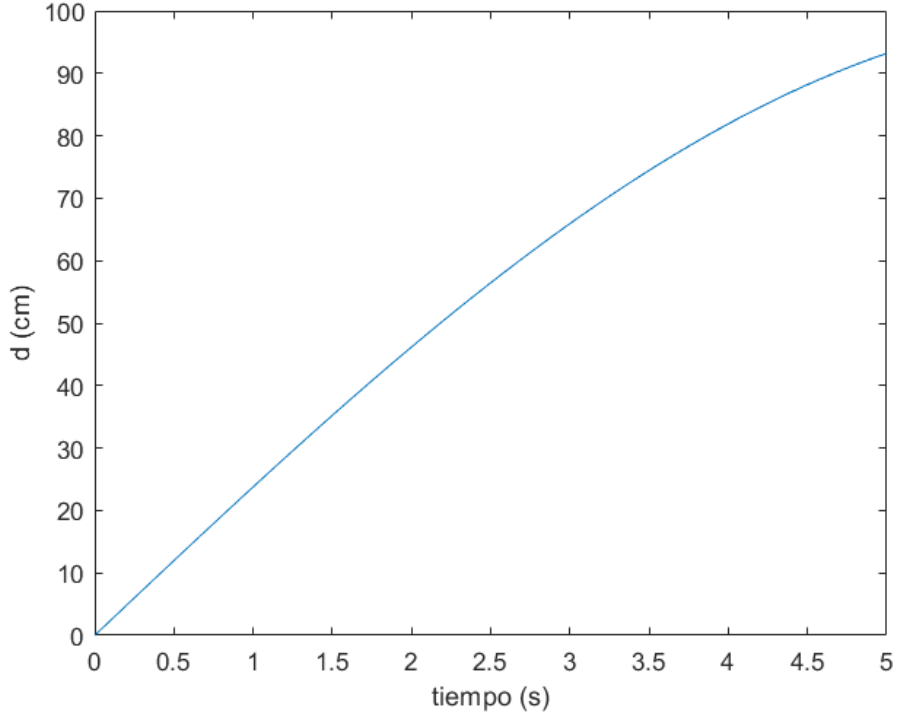

2.4. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DE LA VELOCIDAD DEL CENTRO DE GRAVEDAD Y DE LA COORDENADA ANGULAR DE CADA BARRA

Sistema de ecuaciones que gobierna la velocidad:

$$-r_2\dot{\varphi}_2 sen\varphi_2 - \dot{r}_3 = 0$$
$$r_2\dot{\varphi}_2 \cos\varphi_2 - \dot{d} = 0$$
$$-r_5\dot{\varphi}_5 sen\varphi_5 - \dot{r}_6 = 0$$
$$r_5\dot{\varphi}_5 \cos\varphi_5 + \dot{d} = 0$$
$$\dot{\varphi}_2 - \omega_2 = 0$$

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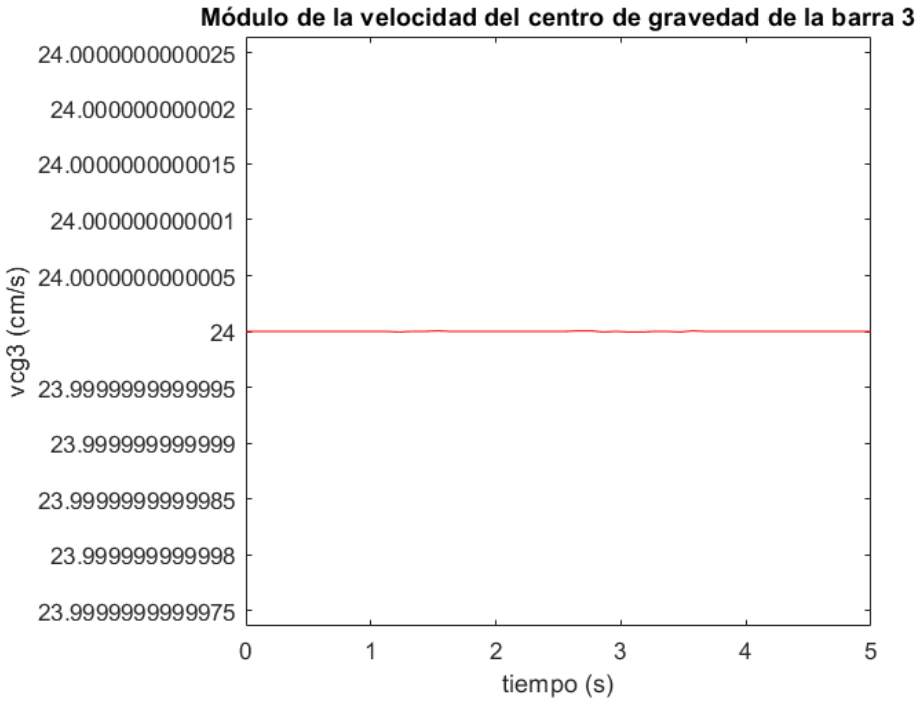

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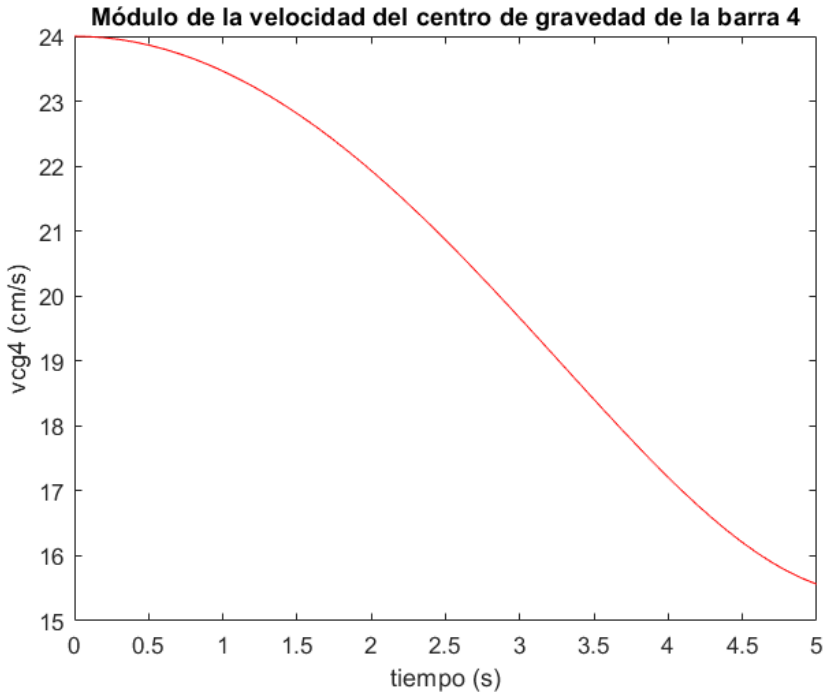

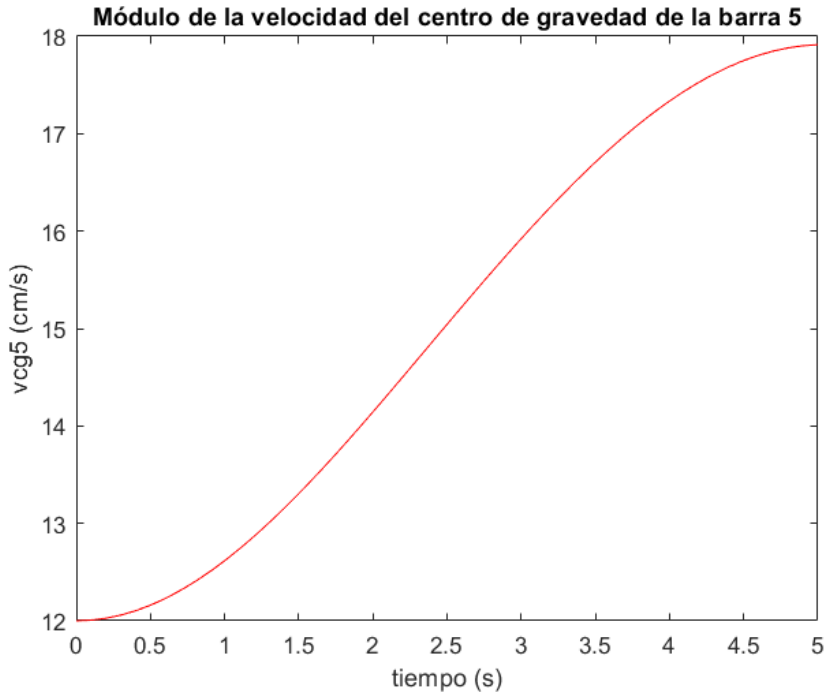

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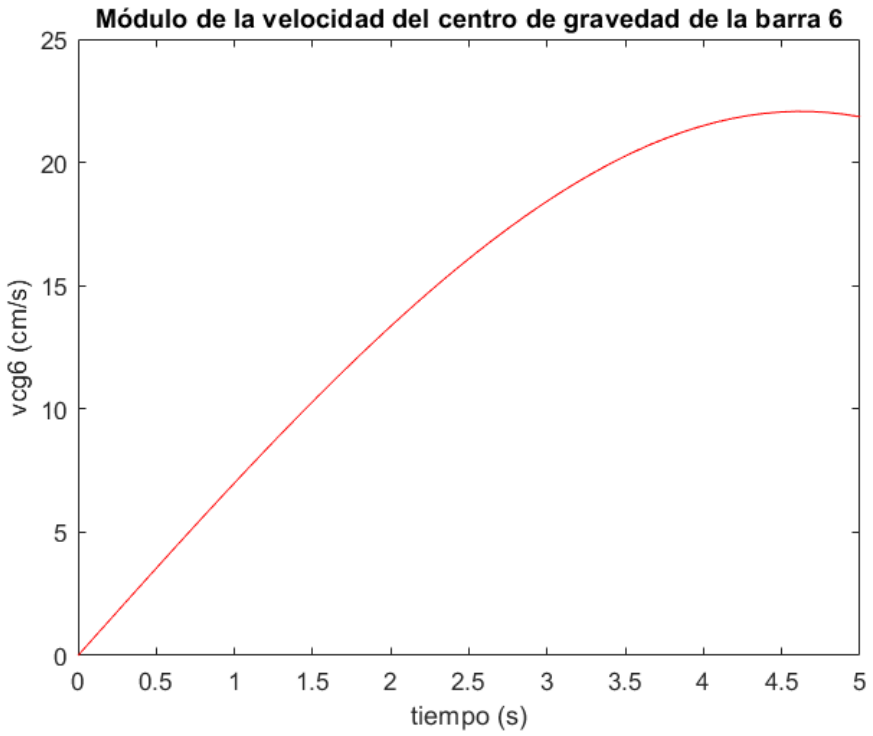

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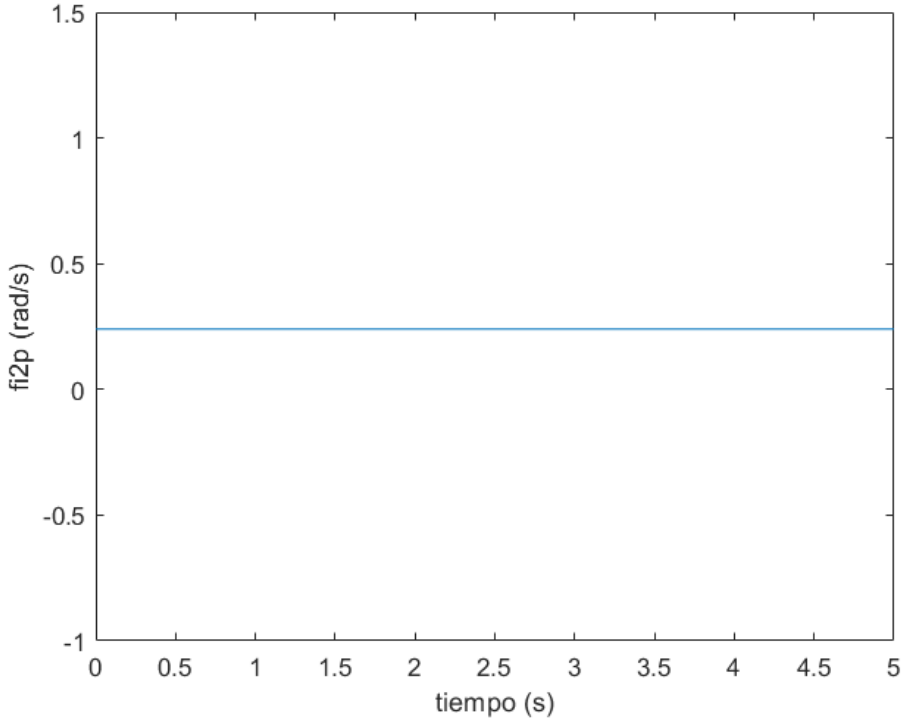

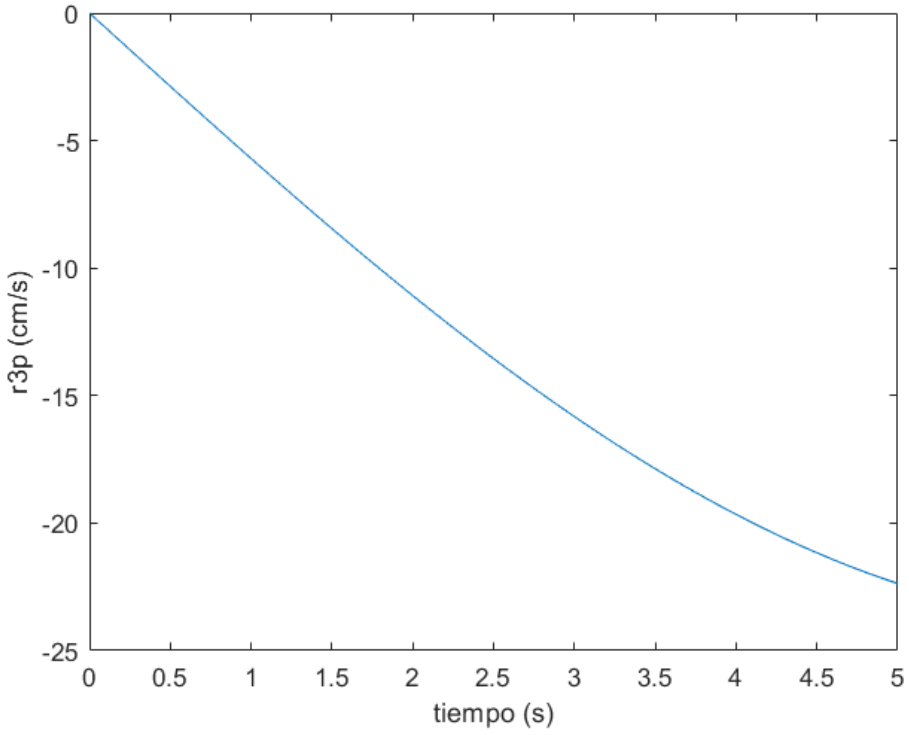

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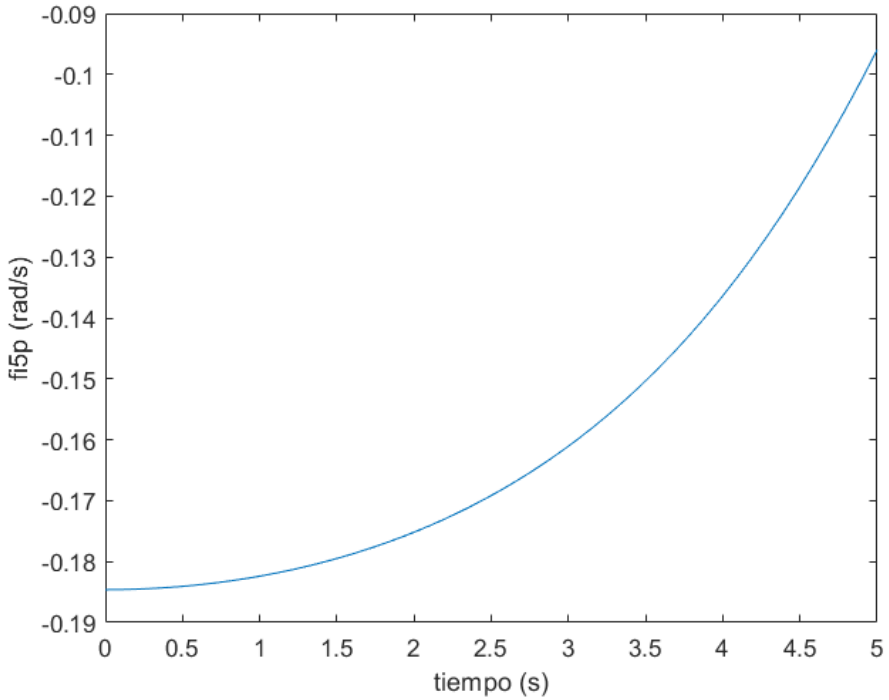

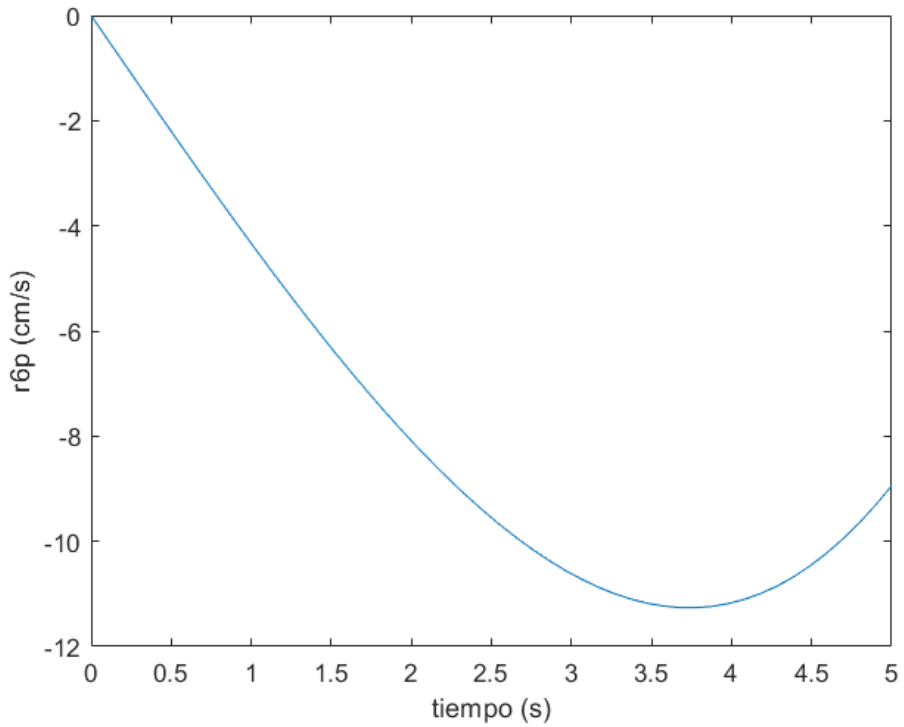

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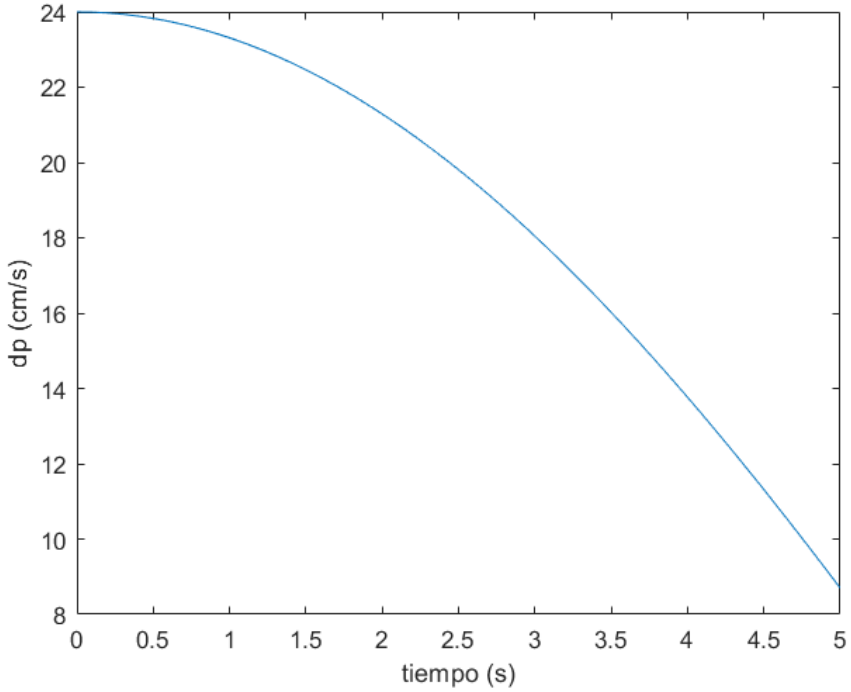

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Sistema de ecuaciones que gobierna la aceleración:

$$-r_2\ddot{\varphi}_2 \operatorname{sen}\varphi_2 - r_2\dot{\varphi}_2{}^2 \cos\varphi_2 - \ddot{r}_3 = 0$$
$$r_2\ddot{\varphi}_2 \cos\varphi_2 - r_2\dot{\varphi}_2{}^2 \operatorname{sen}\varphi_2 - \ddot{d} = 0$$
$$-r_5\ddot{\varphi}_5 \operatorname{sen}\varphi_5 - r_5\dot{\varphi}_5{}^2 \cos\varphi_5 - \ddot{r}_6 = 0$$
$$r_5\ddot{\varphi}_5 \cos\varphi_5 - r_5\dot{\varphi}_5{}^2 sen\varphi_5 + \ddot{d} = 0$$
$$\ddot{\varphi}_2 = 0$$

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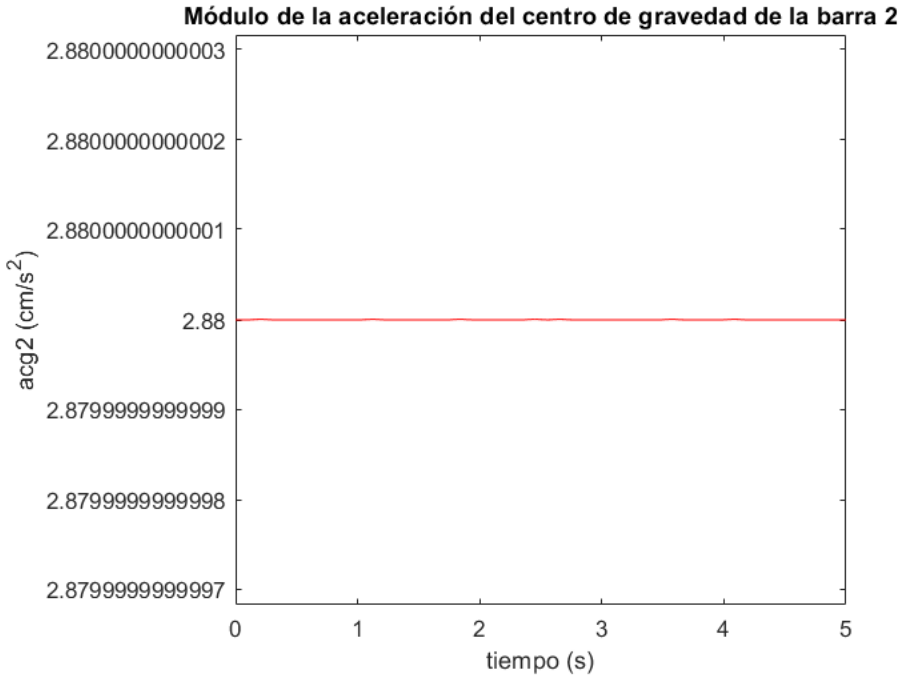

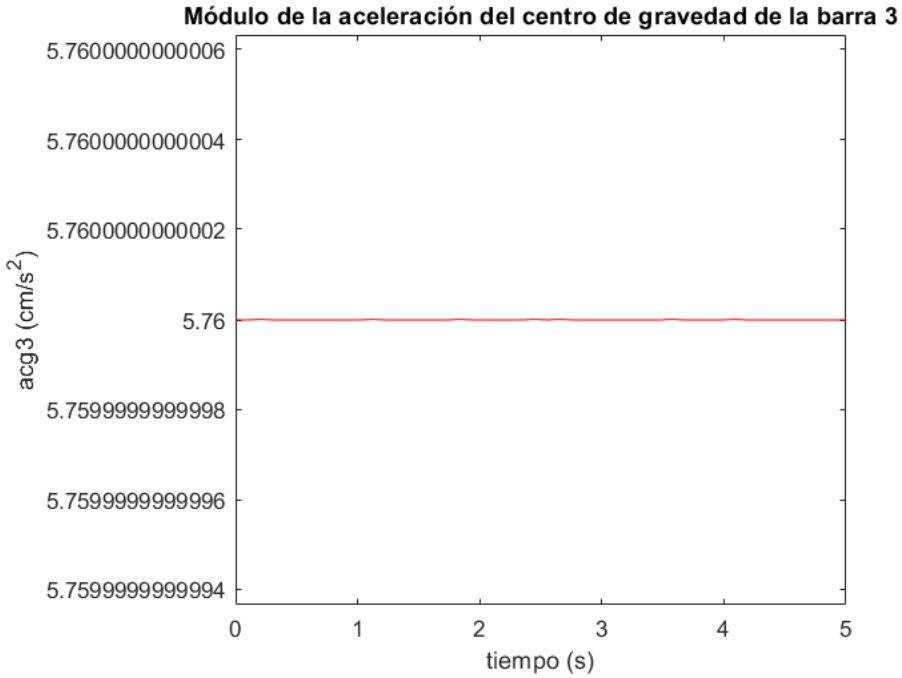

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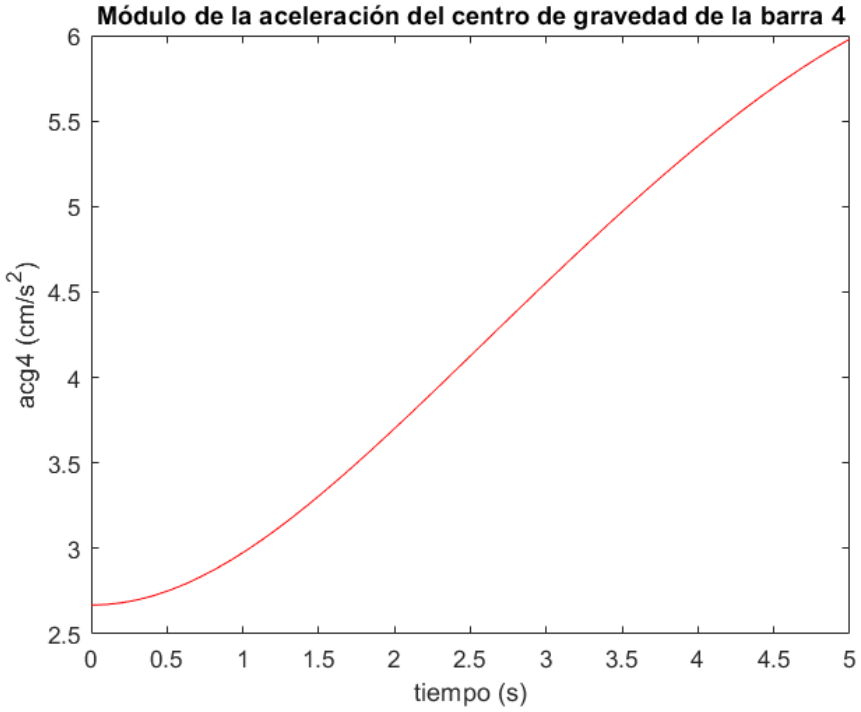

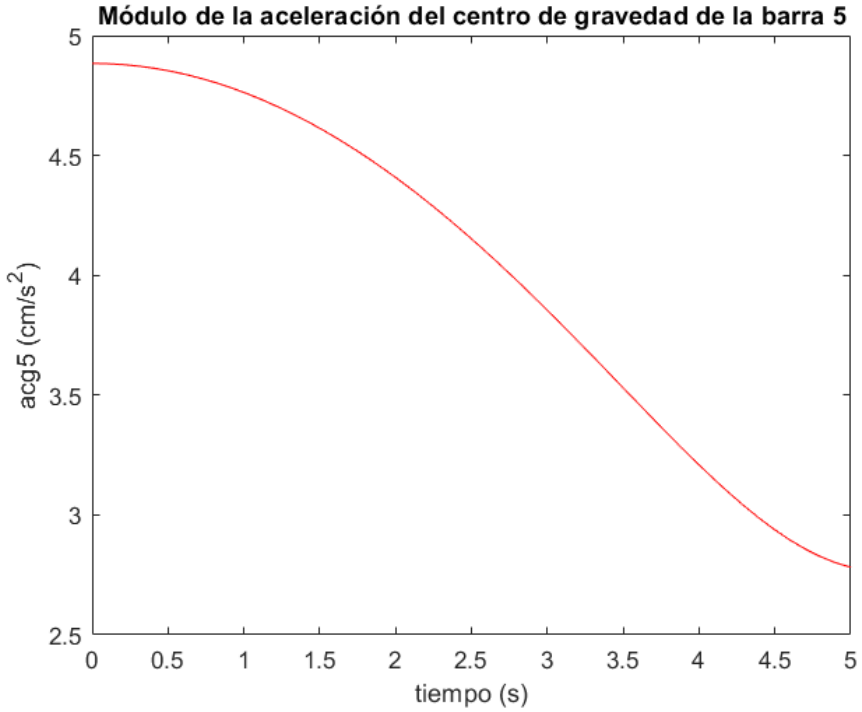

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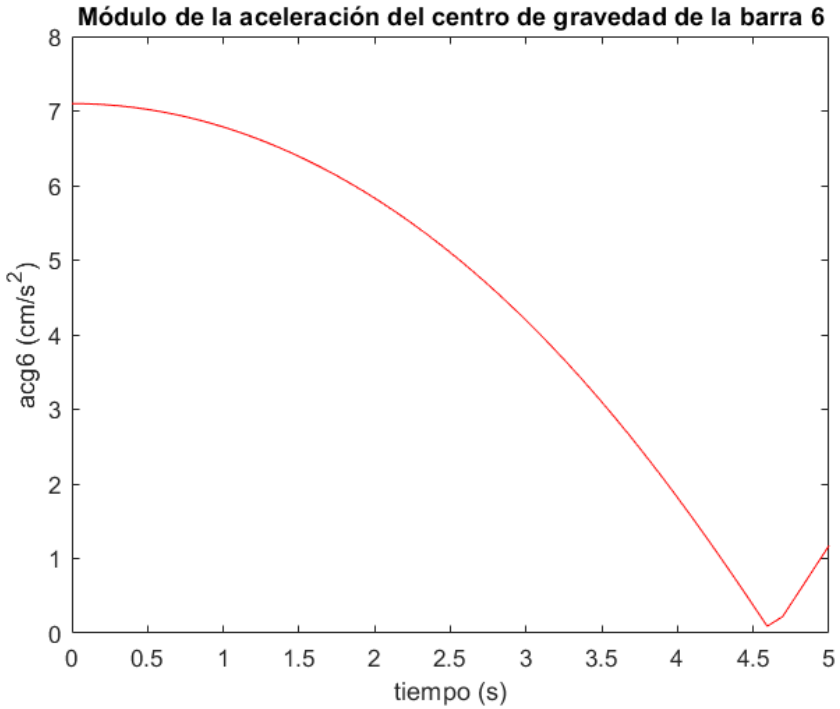

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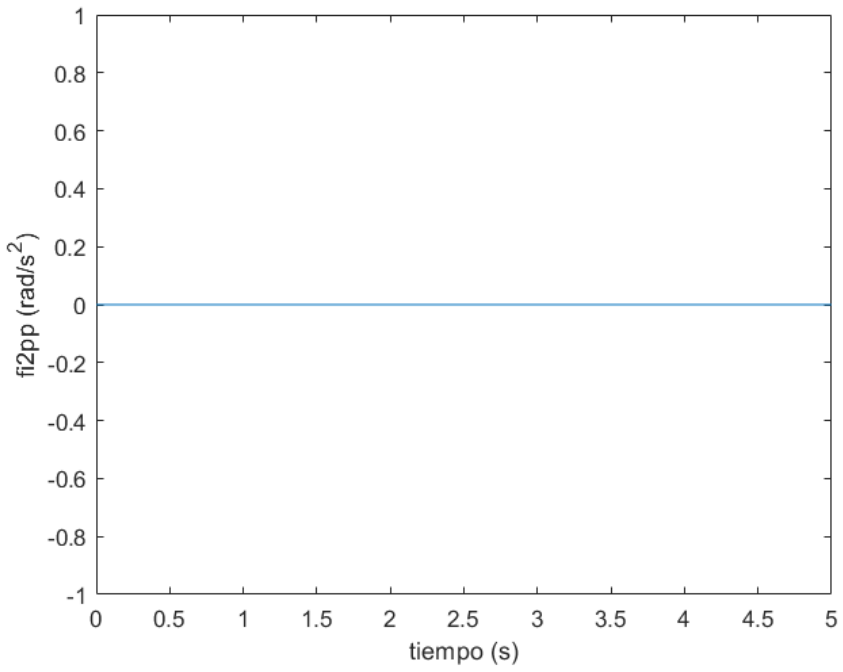

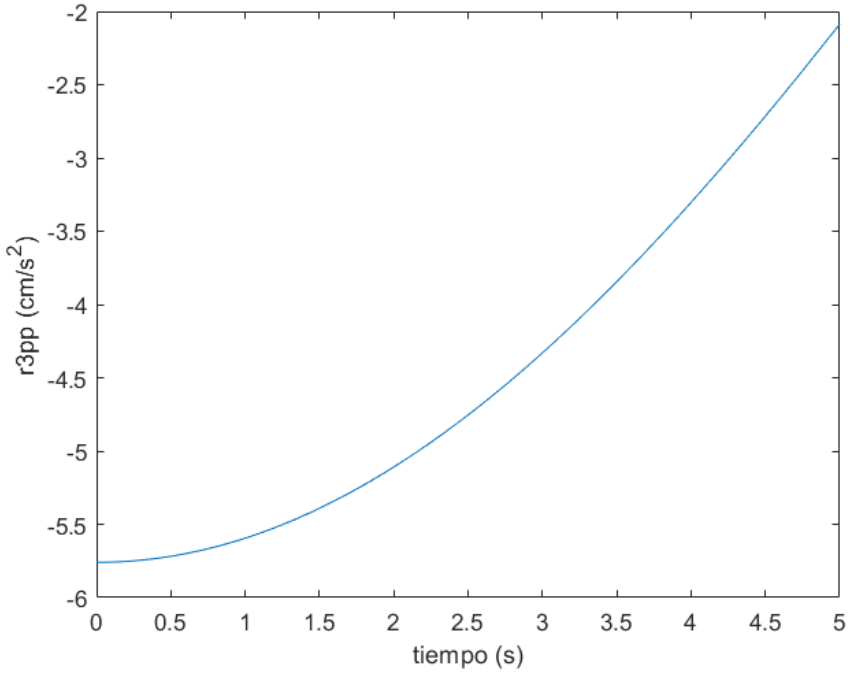

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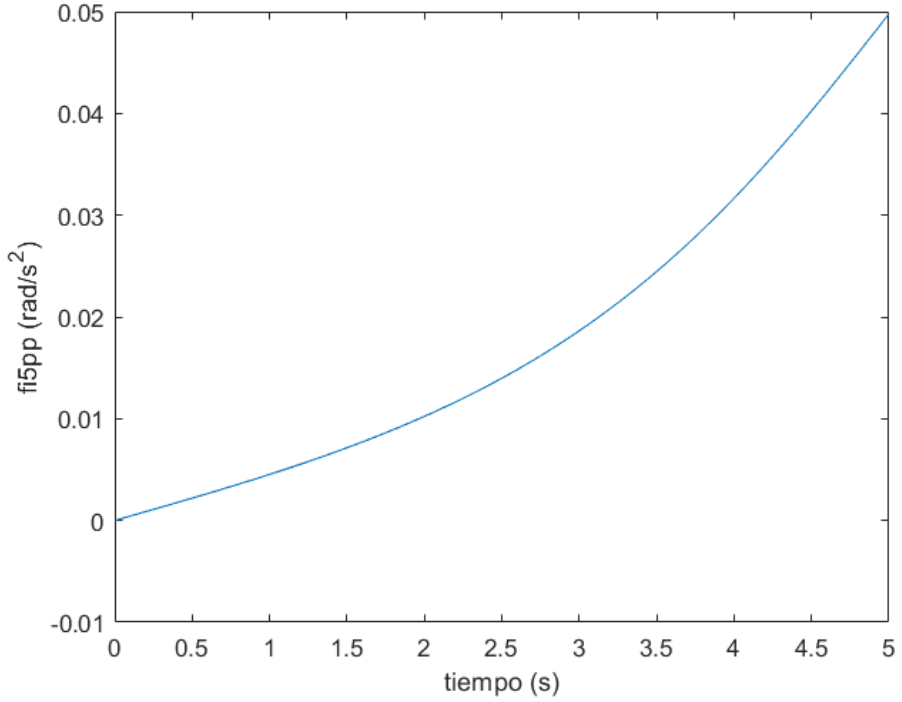

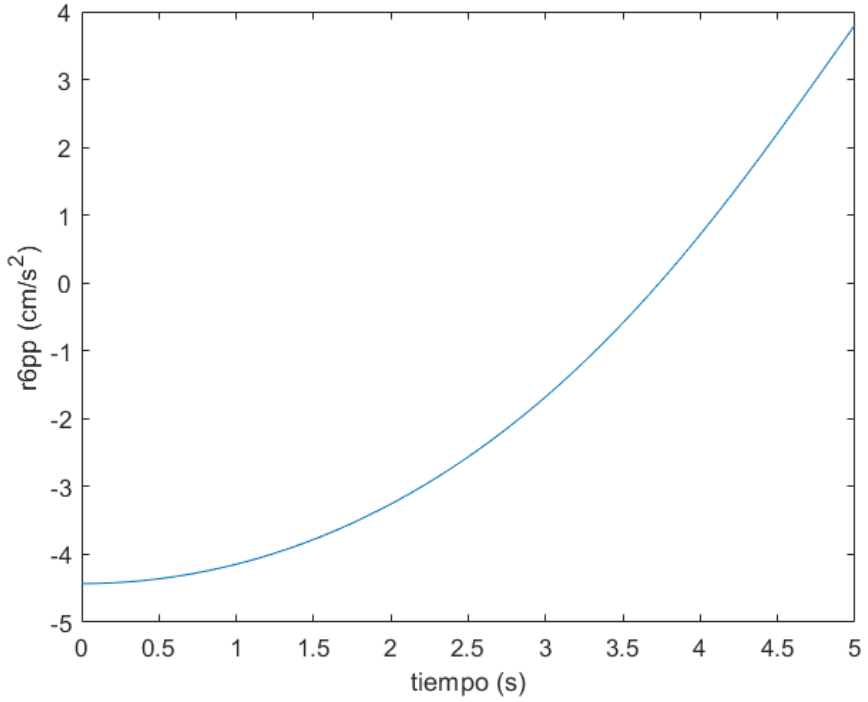

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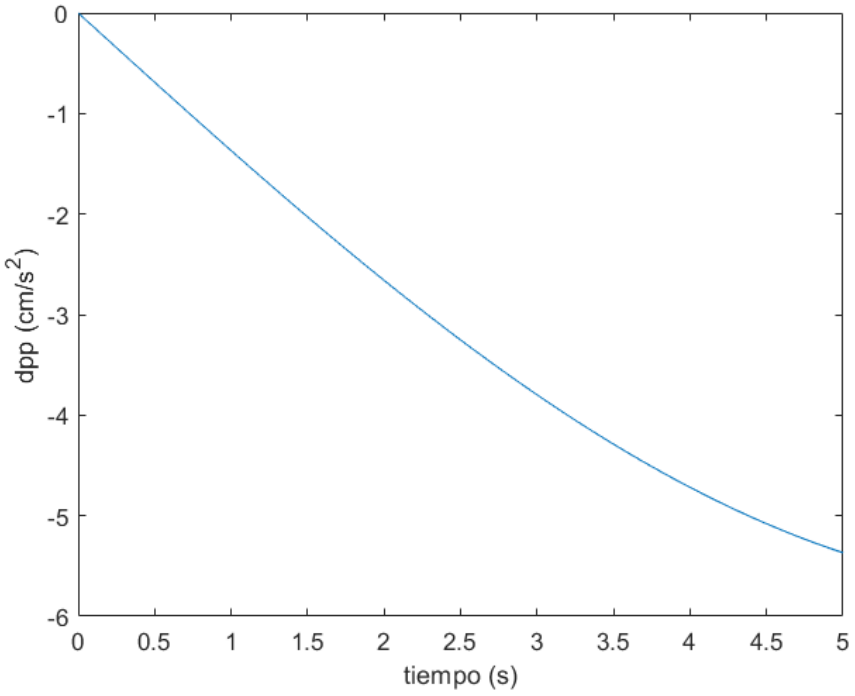

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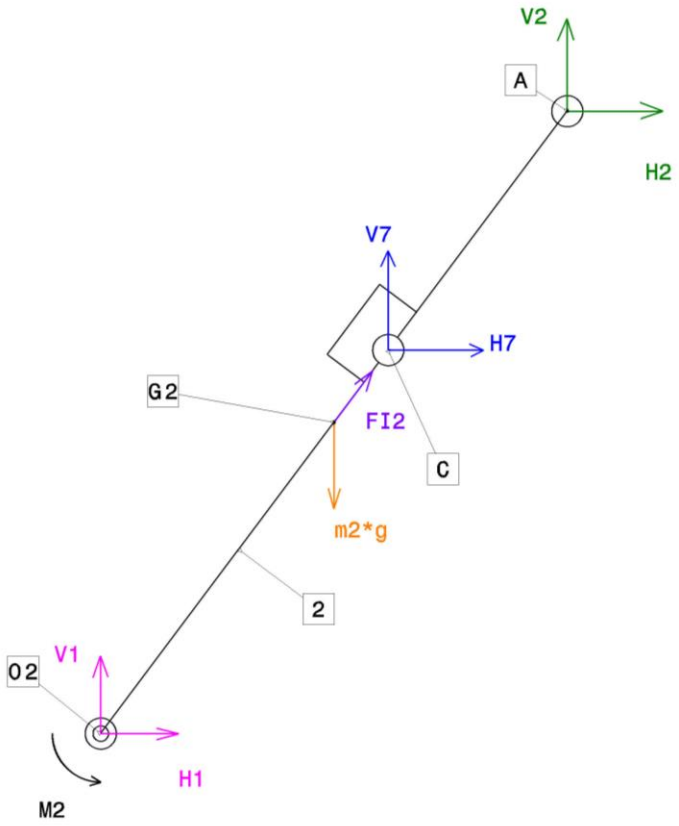
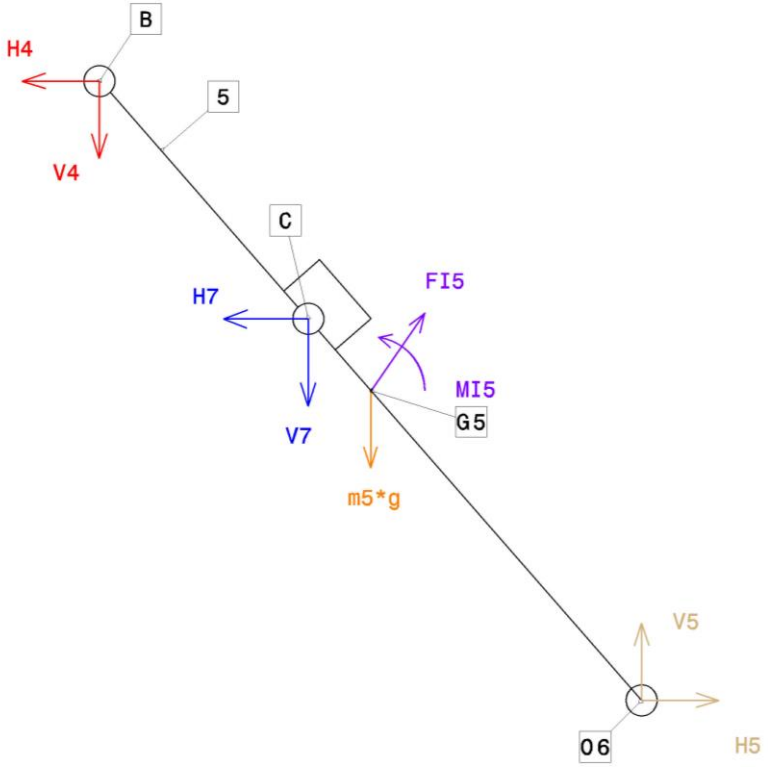
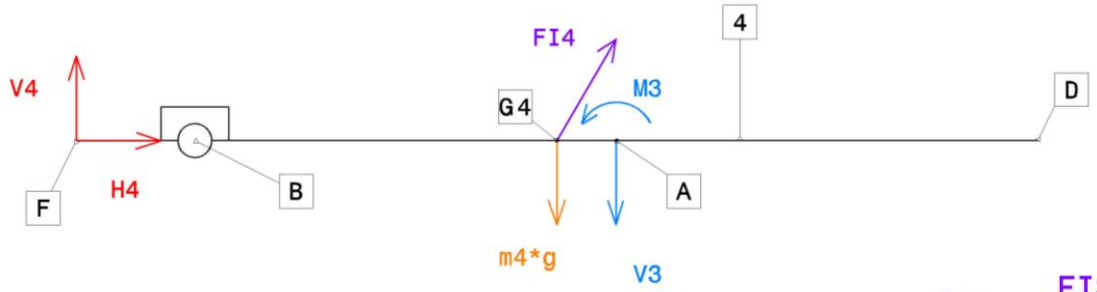
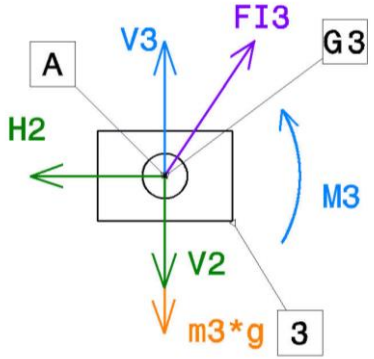
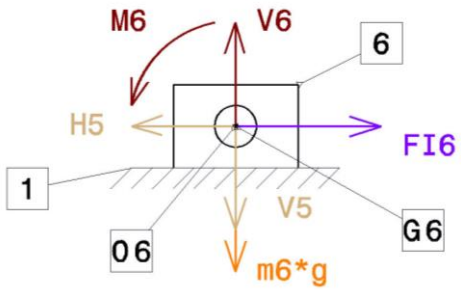










3.1. ECUACIONES DE LA DINÁMICA

$$H_5 = -m_6\ddot{x}_{G6}$$
$$H_4 = m_4\ddot{x}_{G4}$$
$$H_2 = -m_3\ddot{x}_{G3}$$
$$H_7 = -H_4 + H_5 - m_5\ddot{x}_{G5}$$
$$H_1 = m_3\ddot{x}_{G3} - H_7 + m_2\ddot{x}_{G2}$$
$$M_6 = 0$$
$$M_3 = 0$$
$$V_3 = -\frac{(m_4 g + m_4\ddot{y}_{G4})(x_{G4} - x_B)}{x_A - x_B}$$
$$V_4 = V_3 + m_4 g + m_4\ddot{y}_{G4}$$
$$V_2 = V_3 - m_3 g - m_3\ddot{y}_{G3}$$

$$V_7 = \frac{-H_4(y_B - y_{G5}) - H_7(y_C - y_{G5})}{x_{06} - x_C} + \frac{I_{G5}\ddot{\varphi}_5 - (V_4 + m_5 g + m_5\ddot{y}_{G5})(x_{06} - x_{G5})}{x_{06} - x_C} + \frac{-H_5(y_{G5} - y_{06}) - v_4(x_{G5} - x_B)}{x_{06} - x_C}$$

$$V_5 = V_4 + V_7 + m_5 g + m_5\ddot{y}_{G5}$$
$$V_6 = m_6 g + V_5$$
$$V_2 = V_3 - m_3 g - m_3\ddot{y}_{G3}$$

$$\boxed{M_2 = V_1 x_{G2} - H_1 y_{G2} - V_2(x_A - x_{G2}) + H_2(y_A - y_{G2}) - V_7(x_C - x_{G2}) + H_7(y_C - y_{G2})}$$

3.1. ECUACIONES DE LA DINÁMICA

$$T_2 = \frac{1}{2} m_2 v_{G2}^2 + \frac{1}{2} I_{G2} \dot{\varphi}_2^2$$

$$T_3 = \frac{1}{2} m_3 v_{G3}^2$$

$$T_4 = \frac{1}{2} m_4 v_{G4}^2$$

$$T_5 = \frac{1}{2} m_5 v_{G5}^2 + \frac{1}{2} I_{G5} \dot{\varphi}_5^2$$

$$T_6 = \frac{1}{2} m_6 v_{G6}^2$$

$$T = T_2 + T_3 + T_4 + T_5 + T_6$$

$$V_2 = m_2 g \frac{r_2}{2} \operatorname{sen} \varphi_2$$

$$V_3 = m_3 g r_2 \operatorname{sen} \varphi_2$$

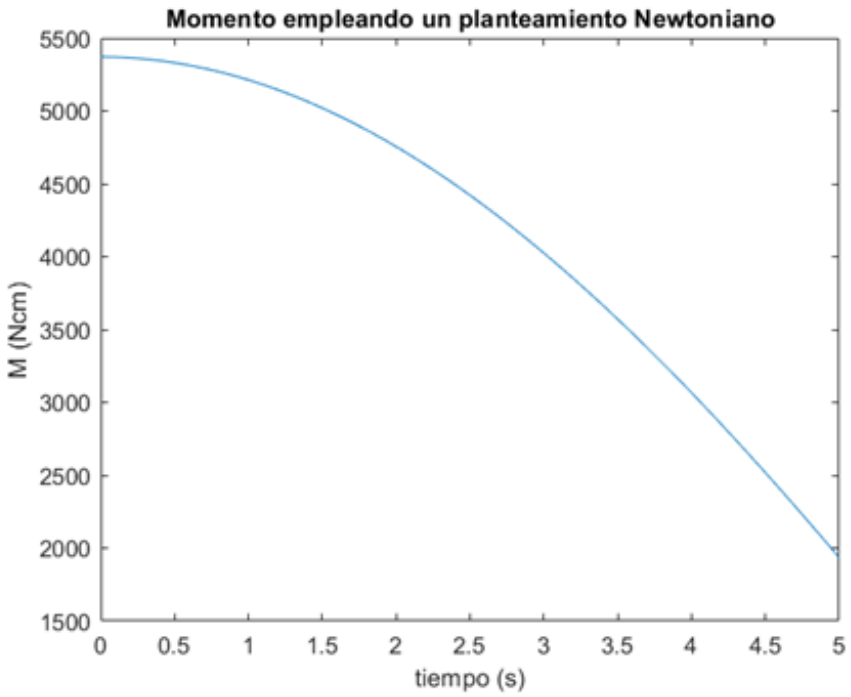
$$V_4 = m_4 g r_2 \operatorname{sen} \varphi_2 \; V_5 = m_5 g \frac{r_2}{2} \operatorname{sen} \varphi_2$$

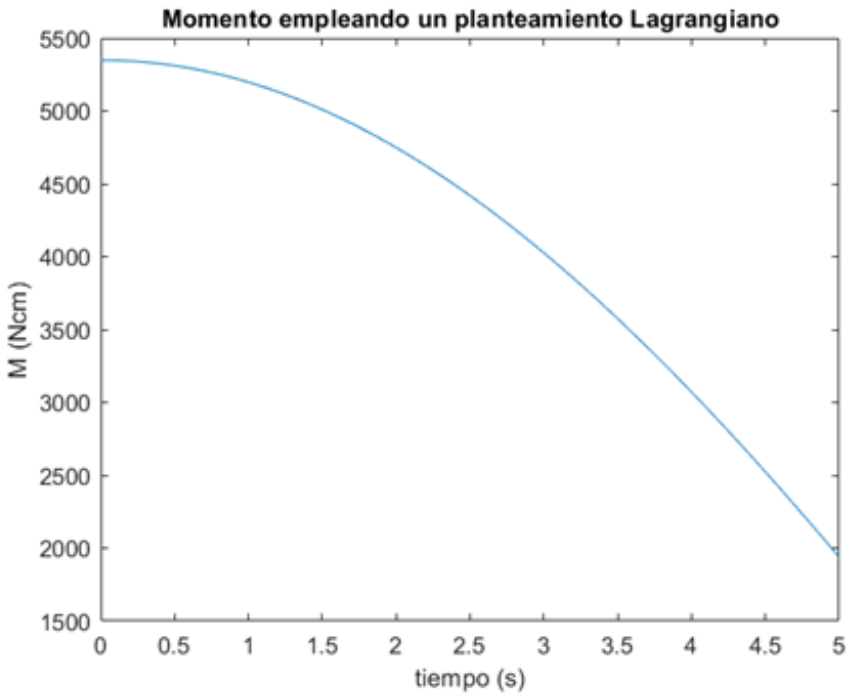
$$V_5 = m_5 g \frac{r_2}{2} \operatorname{sen} \varphi_2$$

$$V = V_2 + V_3 + V_4 + V_5$$

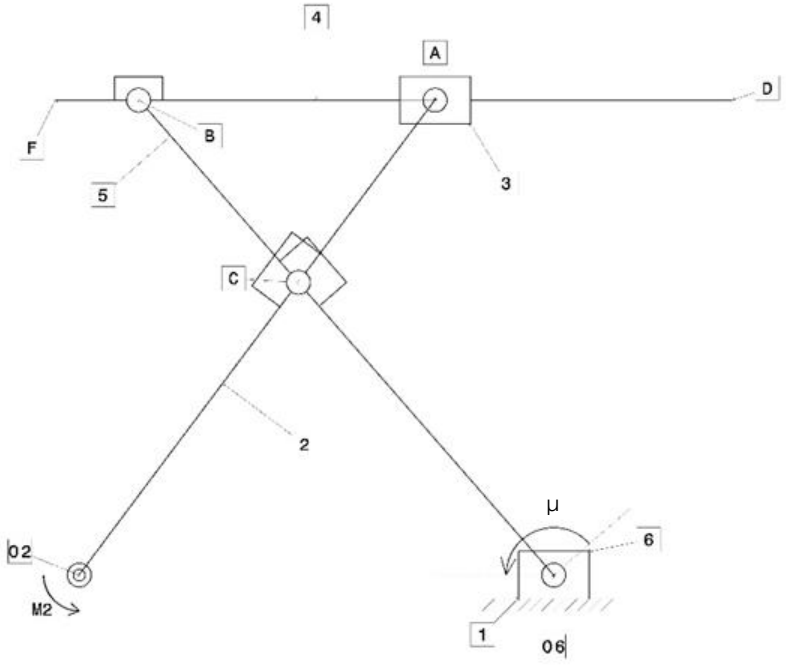
$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\varphi}_2}\right) - \frac{\partial T}{\partial \varphi_2} + \frac{\partial V}{\partial \varphi_2} = M_2$$

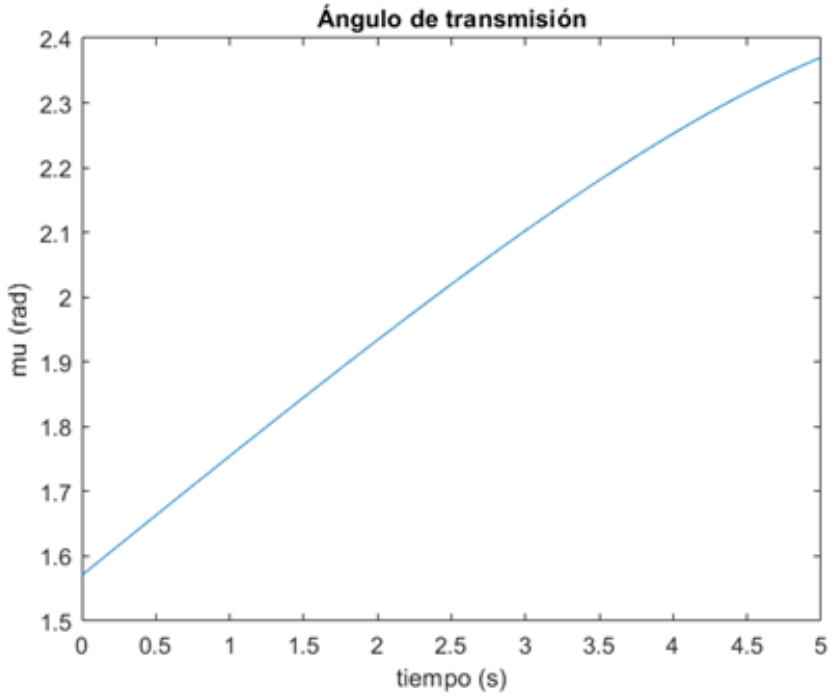
3.2. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DEL MOMENTO RESPONSABLE DEL MOVIMIENTO Y DEL ÁNGULO DE TRANSMISIÓN.



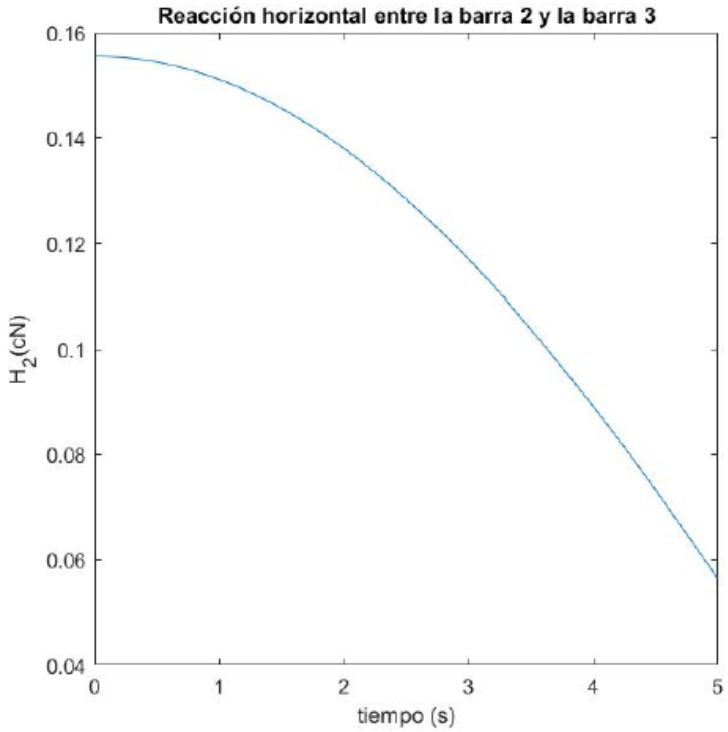


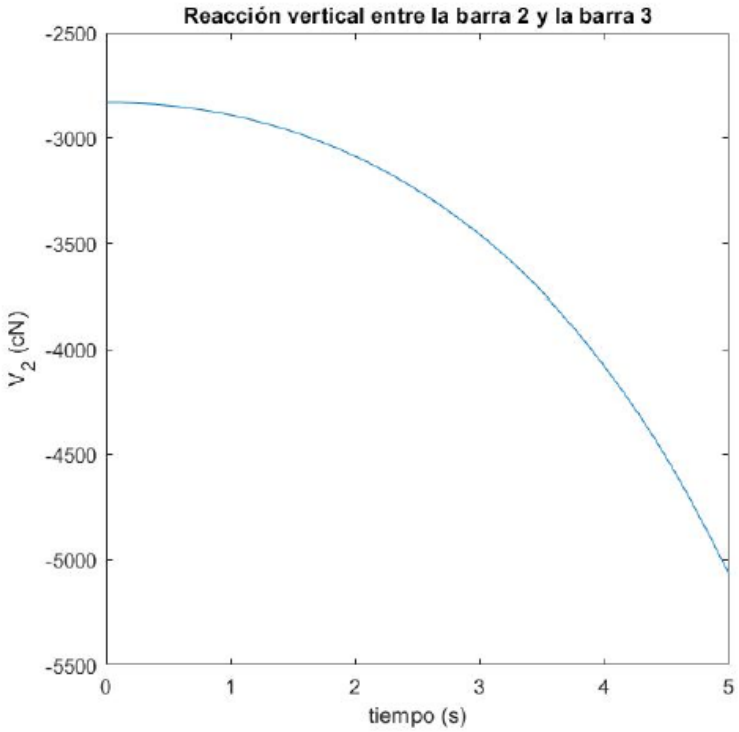
3.2. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DEL MOMENTO RESPONSABLE DEL MOVIMIENTO Y DEL ÁNGULO DE TRANSMISIÓN.

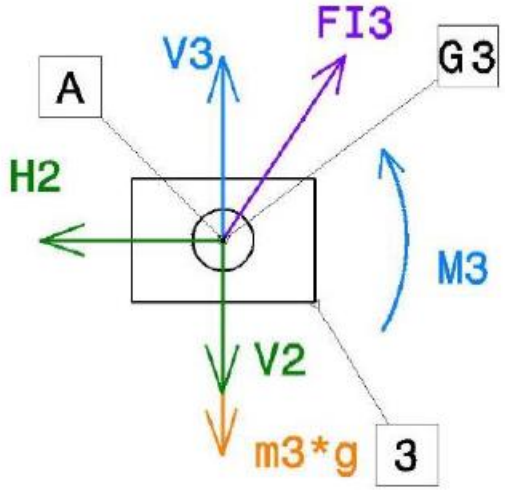




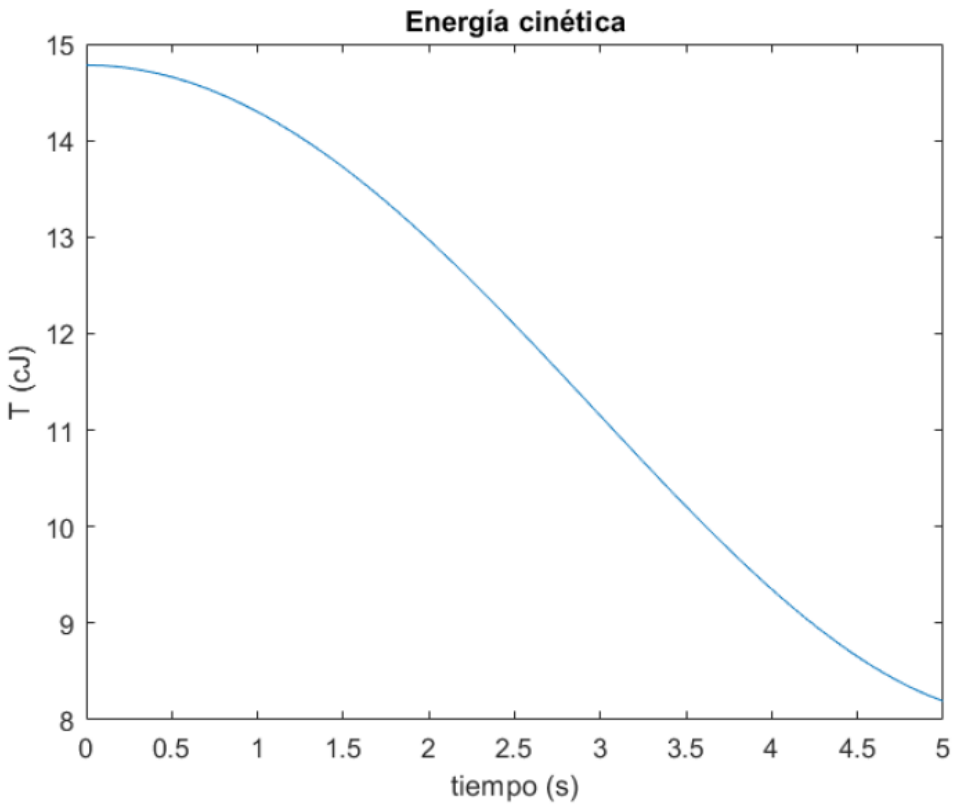
3.3. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DE LAS REACCIONES ENTRE DOS BARRAS

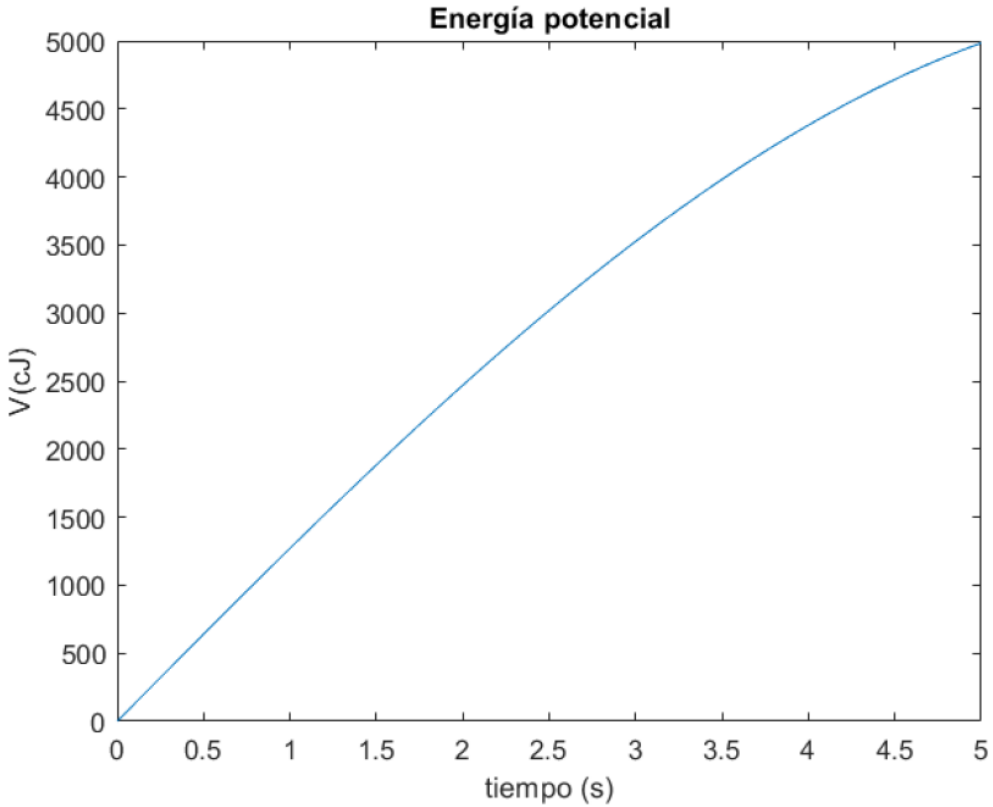




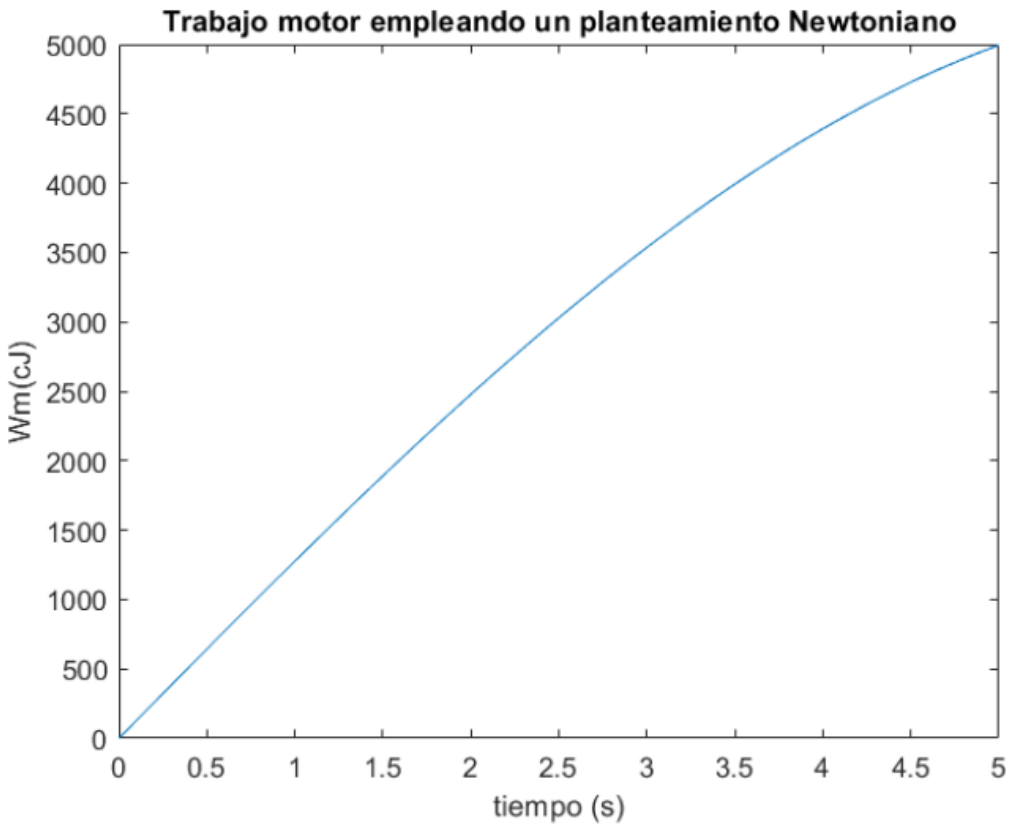


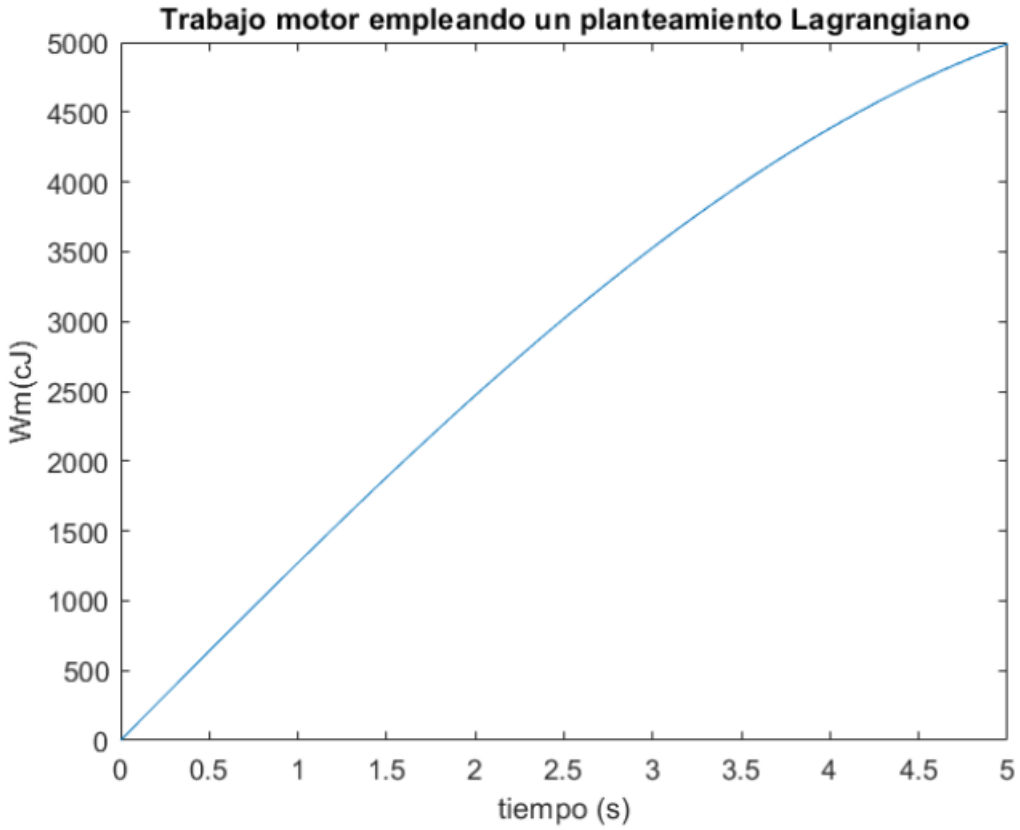
3.4. - 3.5. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DE LAS ENERGÍAS CINÉTICA Y POTENCIAL



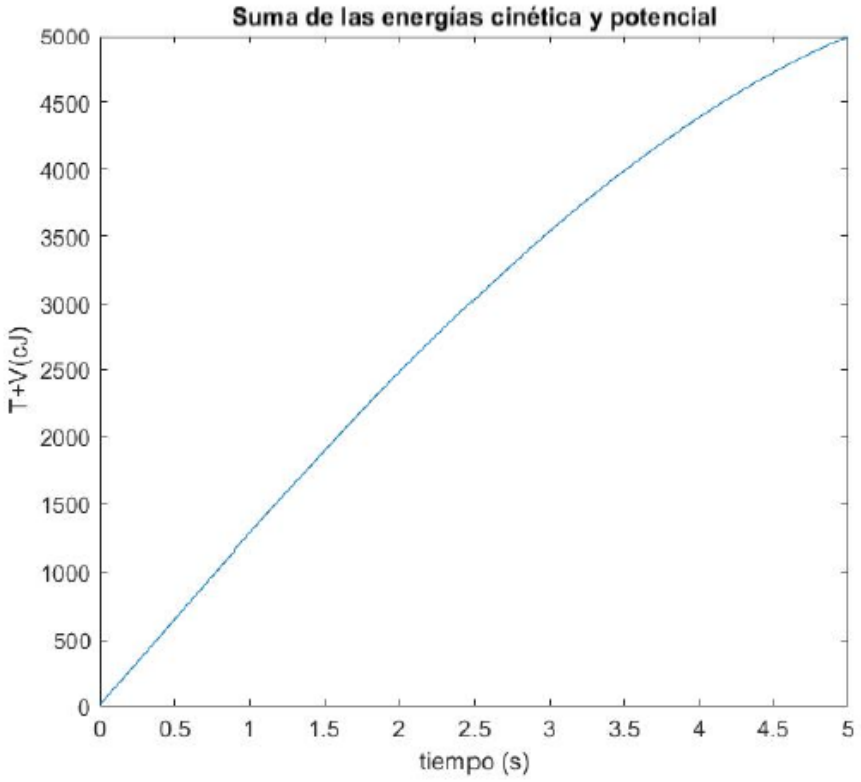


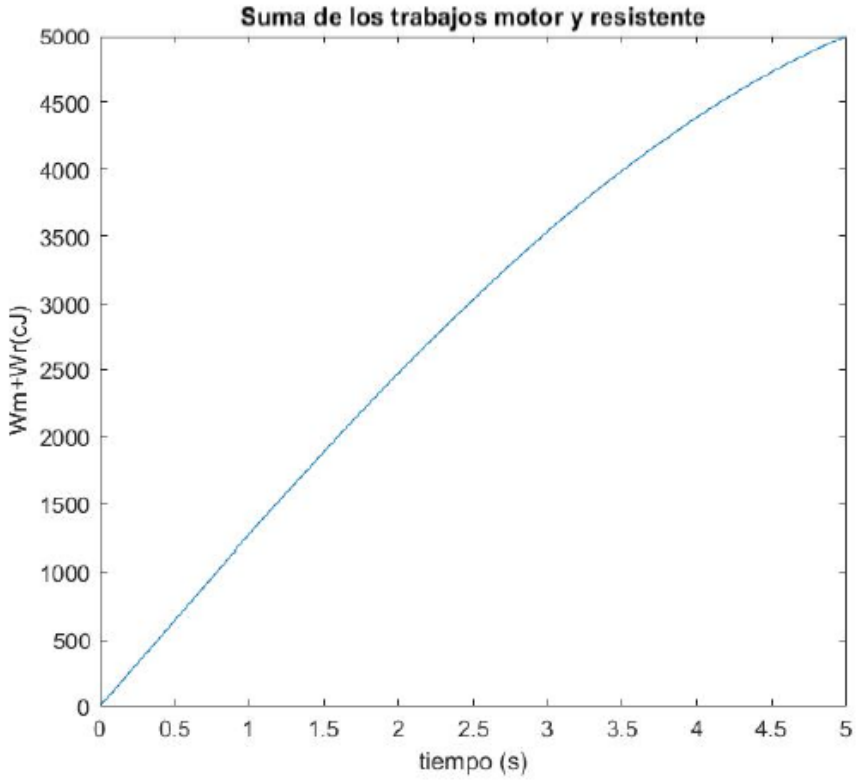
3.6. – 3.7. GRÁFICOS DE LA EVOLUCIÓN DEL TRABAJO MOTOR Y RESISTENTE A LO LARGO DE UN CICLO





3.8. GRÁFICOS DE LA EVOLUCIÓN TEMPORAL DE LA EVOLUCIÓN DE LA SUMA $\Delta T + \Delta V$ Y DE LA SUMA $Wm + Wr$





DUDAS Y PREGUNTAS

FIN